

Tom Stepp

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Experience

Software Engineer, Google Cloud Dataflow	2021 – Present
- Led data-plane design for Dataflow Streaming High Availability: zero data loss, exactly-once processing - Designed persistent streaming shuffle RPCs: eliminated worker OOMs and cut regional fleet peak memory 28% - Optimized Pub/Sub sink and asymmetric autoscaling, cutting customer pipeline costs by up to 70% - Designed delegated Kafka deduplication: 55% lower read cost and 5x better tail latency - Landed sub-sharded state commits on 13,000+ pipelines, cutting delivery latency by up to 50%	
Software Engineer, Boeing AvionX	2017 – 2021
- Architect of continuous integration suite for flight software team of 150 developers - Reduced Matlab code generation times from 10 to 1.5 hours with parallelization and caching	
Software Engineer, Facebook	2016
- Created scripts for automation of network switch testing with Tcl and Ixia hardware API - Developed driver for IC chip with C++ to improve Wedge100 switch function and reliability	
Software Engineer, GE Aviation	2015
Developed test validation software for Leap and Passport aircraft engines	

Skills

C++, Java, Python, GCP Dataflow, Pub/Sub, GCS, Apache Beam, Kafka, Iceberg, distributed systems, stream processing.

Education

M.S. in Computer Science, University of Southern California	2021
B.S. in Electrical Engineering, Purdue University	2017
Study Abroad, Universidad Carlos III de Madrid	2016

Recognition

Apache Beam Summit Speaker: Four talks on Dataflow Streaming performance and scalability	2023 – 2026
Apache Beam College Instructor: Streaming Kafka Pipelines and Scaling Iceberg Ingestion	2025 – 2026
Google Perfy Awards: 1 Gold, 3 Silver, 2 Bronze for Dataflow Streaming performance work	2023 – 2026
Google Healthy Awards: Gold and Silver for code health and build infrastructure migrations	2025 – 2026
Boeing Technical Journal: Analysis for Large-scale Software Systems	2021
Boeing Technical Excellence Conference: Three presentations incl. Model-Based DevOps	2020 – 2021
Boeing Intellectual Property Management: Meritorious Invention Award	2020

Leadership

Mentor & Technical Interviewer, Google	2022 – Present
STEM Outreach Volunteer, Boeing	2018 – 2021
ECE Student Society Board Member, Purdue	2014 – 2017
Eagle Scout, Boy Scouts of America	2013

Projects

ML Handwritten Digit Classification	2020
Programmed neural network from scratch in Python to classify digits from MNIST database	
Go-Playing AI Agent	2020
Programmed Minimax algorithm in Python to beat other AI agents at games of 5x5 Go in real-time	
Solr Search Engine	2021
Created inverted index of news web pages, implemented auto-complete and spelling correction	
Weenix Kernel	2021
Implemented processes and threads, virtual filesystem, and virtual memory	